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Structures Lab Safety Report

Insertion force testing of intraosseous needles

Date:	28/10/2025
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Signatures:

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Supervisor:
(Prof Kristiaan Schreve)

Lab Engineer:
(Karl Zapke) 29/10/25

Head of Safety:
(C Zietsman)

Pressure vessels or pipes (check relevant box):

☒ No pressure vessels or pipes with pressure in excess of 50kPa are involved in this project.

☐ Pressure vessels or pipes in excess of 50kPa are involved – additional signature and report required (refer to Safety Report Guidelines on SUNLearn).

Hot work / working at heights / confined entry / excavation (check relevant box):

☒ No hot work / working heights / confined entry / excavation work involved in this project.

☐ Hot work / working heights / confined entry / excavation work (underline relevant work type(s)) involved in this project – additional signature and report required (refer to Safety Report Guidelines on SUNLearn).

1. Overview of Insertion force Testing

Type of test and standard (if applicable):

Test type	Insertion force test for intraosseous needle using compression test
Standard(s)	—

Equipment to be used:

Equipment type	Make & model	Measurement range (if applicable)	Resolution (if applicable)
Tensile tester	MTS criterion model 44	0 kN – 25 kN	
50 N Load cell	HBM S2M	0-50 N	
Grips	Lab steel grips	<30 kN	
Clamps			
Intraosseous needle	18 Gauge		
Spinal needle	18 Gauge		
IO Trainer			
Data Acquisition System	HBM QuantumX		

Sample geometry:

Sample Cross-Section*	Expected Max. Stress*	Expected Max Load	Number of Samples
		20 N	30

Detailed Experimental Procedure

Introduction

The purpose of this experiment is to determine the insertion force and force profile of an intraosseous needle and a spinal/lumbar puncture needle placed into a model simulating the insertion site, infant tibia. The results of this test will be used to compare the performance of a designed low-cost solution in the form of a new intraosseous needle design or an adaption to a spinal needle to provide better functionality and strength for intraosseous infusion.

Setup Procedure

<u>Power and software</u>

1. Power on the computer and allow the operating system to load fully before proceeding.

2. Activate the MTS Criterion Tensile Tester by switching on the main power using the switch on the right side of the machine and wait for the machine to be in “Standby”-mode.

3. Open the TestWorks 4 software on the computer to access the test control interface.

4. Create my own method in MTS software to do the tensile test I want (Follow the MTS software handbook available in the cabinet)

Safety Checks

1. Check loadshedding schedule for power outages and turn of machines 10 mins. before loadshedding commences.

2. Test emergency stop button

3. Test mechanical interlocks

Connections and loading

1. Install the steel grips in the tensile testing machine using the appropriate pins.

2. Connect the Load cell to the steel grips and tighten the grips.

3. Attach the 3D-printed upper jig to the load cell using the threaded bolt interface. Ensure correct alignment and that the jig is firmly secured.

4. Insert the needle into the 3D printed jig and ensure it is secure.

5. Mount the infant tibia model or IO trainer block onto the lower platen of the testing frame. Use clamps, screws, or a base fixture to ensure the model cannot move or tilt during testing.

Testing

Testing procedure

1. Set the crosshead position to the initial zero point, ensuring the needle is positioned just above (1–2 mm clearance) the tibia model surface.

2. Define crosshead travel limits in the software to prevent over-travel after full penetration.

3. Confirm that the maximum load limit is set below 50 N to avoid overloading the load cell.

4. Lower the crosshead slowly until the needle just contacts the surface of the tibia model.

5. Zero the load and displacement at the point of initial contact.

6. Begin the test run by initiating the programmed method in TestWorks 4.

7. Allow the crosshead to advance at the set speed, driving the IO needle through the tibia model.

8. Stand close to the emergency stop button and be ready to press it in the case of an emergency.

9. Continuously monitor the force–displacement curve on the screen.

10. Stop the test once the needle has fully penetrated the model or when the programmed limit is reached.

<u>Data handling</u>

1. Save the recorded force–displacement data to a designated folder using an appropriate naming convention (e.g., “Sample#_Date_Test#”).

2. Transfer the data to an external drive.

Set down

1. Carefully remove the IO needle from the upper jig.

2. Remove the tibia model from the lower fixture.

3. Detach the 3D-printed jigs and load cell if further testing is not planned.

4. Clean all fixtures and wipe down the test area to remove debris.

5. Power down the MTS machine and computer.

6. Return all components (needle, jigs, manuals) to their designated storage locations.

2. Overview of Silicone Casting

Equipment and materials to be used:

- Speedivac Vacuum Degasser: Vacuum pump and Vacuum chamber
- Dragon Skin Silicone Part A and Part B
- Slacker: Silicone Tactile Mutator
- Mixing bowl
- Mixing stick
- 3D printed mold
- Measuring scale or beakers

PPE required:
- Mask
- Gloves

Detailed Experimental Procedure

Introduction

This procedure involves the fabrication of the part of an intraosseous trainer replicating human skin and flesh of the leg of a paediatric patient. A custom 3D printed plastic mold will be used to shape the silicon. A silicon mix from the supplier AMT Composites will be used to form the IO trainer.

Laboratory Entry Procedure

- Put on all PPE
- Turn the laboratory lights on
- Enter the laboratory using card access

- Turn the extraction fan on
- Set up the workstation for mixing and pouring of the silicone

Fabrication Procedure

1. Mixing

- Mix equal amounts of Dragon Skin Part B with Slacker using a volume or weight measurement and stir thoroughly.
- Mix the same amount of Dragon Skin Part A as Dragon skin Part B with mixture of Dragon Skin Part B and Slacker and stir thoroughly.

2. Degassing

- Degas the mixture in the vacuum degasser for approximately 5 minutes, ensuring the container is at least twice the mixture's volume to avoid spillage.

3. Assemble the 3D printed mold

- Assemble the different pieces of 3D printed mold using bolts and nuts to fasten the assembly.

4. Pouring

- Spray the plastic mold with a non-stick spray.
- Pour the degassed mixture into the mold.

5. Curing

- Place the mold containing the silicon in a safe space to cure at room temperature.
- Leave the mold until the silicone has cured.

6. Remove silicone from mold

- Loosen the nuts and bolts on the plastic mold.
- Slowly remove the silicone part from the plastic mold.

Exit Procedure

- Ensure vacuum degasser is switched off at the plug point.
- Ensure extractor fan is switched off.
- Ensure workstation is cleaned up.
- Ensure the laboratory door is closed.
- Switch off all lights.

3. Warning Symbols

(a) Electrical hazard

(b) Heavy object

(c) General warning

(d) Pinching surfaces

(e) Toxic Fumes

Figure 1: Applicable warning/hazard symbols

4. General Laboratory Safety

The following general laboratory safety instructions are applicable:

- No afterhours testing may be performed without the necessary permissions[1].
- Full supervised training is required before testing may be undertaken. Permission to proceed with unsupervised testing should be signed off by the lab engineer.
- Closed shoes must be worn at all times.
- Emergency equipment must be located and easily accessible.
- Students may not work alone in the laboratory.
- Emergency exits must be known. The nearest exits applicable to the setup are provided in Appendix A of this document.
- Loose clothing may not be worn. Loose hair must be tied up.
- Good housekeeping practices should be maintained during testing. The lab should be completely clean, including all equipment stored away, after testing. Refer to the General Housekeeping section for particulars regarding practices to be followed for this specific setup.
- No food or drink is permitted in the laboratory.
- Safety report must be visible and accessible during testing.
- No equipment or test may be left unattended.

5. Anticipated Interactions with other Laboratory Users

The laboratory space will be shared with other students and experimental setups. Interactions between different setups and other lab users will be minimised as far as possible. Sufficient space will be kept between different experimental setups. There will be efficient communication to other students regarding the times when experiments will take place to ensure no student is interrupted or bothered. In the case where assistance is needed, a laboratory technician can be asked.

6. General Housekeeping

- Return all tools or equipment taken from the mechanical stoor.
- Ensure that electric and electronic equipment is switched off after use.
- Store all clamps, grips and jigs in the appropriate storage location.

[1] Written permission from supervisor and approved by the chief safety officer. Attach proof of this permission to this document and note here the times and conditions of the arrangement.

- Ensure that the workspace is clean, neat and safe.
- Report any unusual activity or broken tools/components.
- Don’t let anyone without access into the labs.
- Don’t leave any personal belongings in the labs.
- Ensure that all cabinets and containers are closed and locked before exiting the lab.

7. Fire Safety

There is no significant fire hazards anticipated for this experiment. The equipment used during this experiment will operate using standard laboratory electrical supply and all materials tested; including needles and IO trainer block are non-flammable. The fire risks are limited to general laboratory electrical hazards. These risks are mitigated by proper and frequent equipment maintenance to ensure all cabling are intact and fire extinguishers are available in the laboratory. All equipment will be switched off after use to further lower the fire risk. Note will be taken on where all fire extinguishers are located in and around the laboratory.

8. Activity Based Risk Assessment

Activity	Risk	Risk Type* (P/E)	Classification of Risk Severity	Mitigating Steps
General				
Entering the lab	Injuries to hands from the gates	P	Acceptable	Use caution when using the gates
Moving around the lab	Tripping or knocking equipment over	P	Acceptable	Never use a cell phone when moving around and be aware of surroundings
Power outages	Damage to equipment or loss of data	E	Possible	Be aware of load shedding schedules and be sure to switch all equipment
Personal valuables in the laboratory	Theft of valuables from the laboratory.	P	Acceptable	Do not bring unnecessary valuable items to laboratory sessions. Valuables that are brought to the laboratory should be placed in a safe and visual location
Tidying lab	Tripping	P/E	Possible	Do not trip over cables
	Cuts	P	Possible	Be aware of sharp edges on equipment or assembly.
Turning on equipment	Electrical shock	P	Possible	Check that all cables are properly connected and the insolation remains intact.

MTS use				
Setting up and taking down the test setup on the machine	Dropping of grips	P/E	Possible	Due to heavy weight, grips should be handled carefully when inserting to machine. Wear appropriate PPE (closed shoes)
	Pinching fingers in grips	P	Possible	Exercise caution to ensure hands and fingers are kept clear of pinch points or areas beneath components during installation. Obtain assistance when lifting or positioning heavy components to prevent injury.
	Crushing limbs in the machine	P	Possible	Keep all limbs clear of the test setup while the crosshead is in motion. Engage the crosshead lock before making any adjustments to the setup. Ensure that no crosshead movement is initiated via the computer or control software while setup adjustments are being performed.
	Damaging the load cell	E	Substantial	Ensure the rail is positioned symmetrically around the load point before starting the test. Move the crosshead to a safe position before beginning assembly. Take care to avoid contact between any components and the load cell during assembly.
Performing Test	Damaging the load cell	E	Substantial	Ensure the load limits are set correctly.
	Crushing limbs in the machine	P	Possible	Stand away from the machine during testing. Lock the crosshead while adjusting the test setup. Ensure no one moves the crosshead using the

				computer while the setup is adjusted.
	Potential for injury caused by component failure resulting in parts being ejected from the test setup.	P	Substantial	Slow and controlled load application limits possibility of parts flying off. Wear safety glasses. Stand away from the machine while testing.
Insertion or removing of the needle from jig.	Needle prick injury	P	Possible	Place needle cap onto needle before handling it.
Silicone Casting				
Cutting components and sharp edges .	Hand injuries	P	Acceptable	Cut away from your body and handle sharp edges with caution
Measuring out silicone	Spillage	P/E	Acceptable	Place a catchment tray or protective cover under the container to prevent damage in case of spillage.
	Incorrect amounts mixed	E	Acceptable	Use proper measurement equations and ensure the scale is zeroed between measurements for accuracy.
	Incompatibility with previous mix	E	Acceptable	Use only clean mixing containers designated for your own use to prevent contamination.
Working with the vacuum chamber	Equipment failure	P/E	Possible risk	Follow all operational procedures and safety instructions provided during the induction.
Degassing	Spillage	E	Acceptable	Use a mixing container at least twice the volume of the silicone to minimize spillage due to gas expansion. Allow the mixture to cure before peeling off any spills — do not wipe liquid silicone.
Removing silicone from mold	Silicone getting stuck in mold	E	Acceptable	Spray the internal surface of the mold with non-stick spray before pouring in the silicon

*P – personal, E - equipment

9. Disciplinary Actions

Failure to comply with any of the aforementioned safety regulations or procedures will result in disciplinary action. Students will be issued an initial warning: after three warnings, the lab access is revoked for a month.

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Appendix A: Emergency Evacuation Plans Structures Lab